

Machine Learning Models for Mental-Health Prediction from Actigraphy: A Systematic Appraisal and an Empirical Demonstration

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Code, run manifests and generated tables: <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal>

Abstract

Background and objective. The open actigraphy cohorts DEPRESJON (depression), PSYKOSE (schizophrenia) and HYPERAKTIV (ADHD) are the most used public benchmarks for machine-learning detection of mental disorders from wearables, and reported accuracies have risen from the dataset authors' leave-one-patient-out 0.7 to above 0.9. We asked whether this reflects better models or weaker evaluation. **Methods.** Part I appraises every published supervised model on the cohorts with PROBAST+AI and TRIPOD+AI, recording the unit of train/test splitting, external validation, calibration reporting, events per predictor, code availability and handling of the control group shared by DEPRESJON and PSYKOSE. Part II re-evaluates the cohorts with models fixed a priori, ten seeds and participant-level bootstrap intervals: record-wise versus subject-wise cross-validation (E1), frozen cross-cohort transfer (E2), calibration and recalibration (E3), sex strata (E4), transdiagnostic and five-class tasks (E5) and ablation (E6). **Results.** Of 28 published models, 7 split by participant, 9 by day and 12 did not state the unit; none reported calibration or validated on a second cohort. Record-wise splitting inflated AUROC by 0.065 to 0.114 (DEPRESJON), 0.020 to 0.044 (PSYKOSE) and 0.130 to 0.169 (HYPERAKTIV), where the honest estimate was at chance. Models trained on PSYKOSE lost 0.13 to 0.21 AUROC on DEPRESJON with calibration slopes of 0.27 to 0.63; honest slopes were 0.53 to 0.95, and recalibration did not repair the transfer. Honest performance was stable under ablation. **Conclusions.** Reported progress on these benchmarks is largely an artefact of evaluation design. We provide subject-wise, calibrated, cross-cohort baselines with public code against which future claims can be checked.

Keywords: actigraphy; depression; schizophrenia; ADHD; DEPRESJON; PSYKOSE; HYPERAKTIV; OBF-Psychiatric; data leakage; external validation; calibration; PROBAST+AI; TRIPOD+AI; reproducibility.

1 Introduction

1.1 Background: prediction models on wearable activity data

A clinical prediction model takes measurements from an individual and returns a probability, or a class label, for a health state. In mental health the measurements increasingly come from wearables. *Actigraphy* is the simplest of these: a wrist-worn accelerometer counts movement once a minute, for days or weeks, without any action by the wearer. The resulting series carries information that clinicians have long associated with mental disorders: lower and flatter daytime activity and disrupted sleep in depression, blunted and fragmented circadian rhythm in psychosis, and restlessness in attention-deficit/hyperactivity disorder (ADHD). Machine learning (ML) offers a way to turn that series into a screening score, and the appeal for AI researchers is obvious: the data are dense, objective, cheap, and publicly available.

Three open datasets from Haukeland University Hospital and collaborating Norwegian sites made this possible. DEPRESJON contains 23 in- and outpatients with unipolar or bipolar depression and 32 healthy controls [1]; PSYKOSE contains 22 inpatients with schizophrenia and the same 32 controls [2]; HYPERAKTIV contains adults referred for ADHD assessment, of whom some received the diagnosis and the rest serve as clinical controls [3]. All were recorded with one device family at one-minute resolution and were re-released in 2025 as the harmonised OBF-Psychiatric collection of 162 participants [4]. They have become the default benchmarks for detecting mental disorders from wearables, and the numbers reported on them have climbed: the dataset authors' leave-one-patient-out F1 of 0.73 for depression [1] has been followed by accuracies of 0.85 to 0.96 [5, 6, 7] and, for schizophrenia, of 0.92 to 0.94 [8, 9]. Read at face value, that is a field approaching clinical usefulness.

1.2 The evaluation problem, stated for a general reader

Whether a reported number means what it appears to mean depends on how the model was tested, and three aspects of testing decide it here.

Which unit is held out. Each participant contributes many days of data, and the days from one person resemble each other far more than they resemble anyone else's: wake time, weekday routine, overall activity level and the pattern of rest are personal signatures. If the days are shuffled into cross-validation folds without regard to who they belong to (*record-wise* or day-level splitting), a flexible model sees other days from the same person in training and can recognise the person rather than the disorder. The estimate that results is not an estimate of performance on a new patient. Holding out whole participants (*subject-wise* splitting) removes the shortcut. The gap between the two is a form of data leakage; it is documented in wearable, EEG and neuroimaging research [10, 11], but it has not been measured on these cohorts, and many papers do not say which unit they used.

Whether the model was tested on a second population. A model that performs well on the people it was developed on may not transfer to a different clinic, a different diagnostic group or a different severity range. Applying a frozen model to an independent cohort, called external validation, is the standard test of that transfer, and it is rarely done. On these cohorts there is an added trap: because PSYKOSE reuses DEPRESJON's control participants, a study that trains on one and tests on the other, or pools them, has scored part of its test set on people it trained on unless it noticed the overlap.

Whether the probabilities are calibrated. Discrimination (AUROC, accuracy) measures whether cases are ranked above controls. Calibration measures whether a predicted probability of 0.8 corresponds to about 80 % of such patients actually having the condition. A screening tool is used through its probabilities, and a miscalibrated model gives confident advice that is wrong in a systematic direction [12]. Calibration is essentially never reported for wearable-based mental-health models.

These three requirements are not new to clinical prediction research. The PROBAST+AI tool [13] scores a study's risk of bias on exactly these points, and the TRIPOD+AI statement [14] lists what an ML prediction study must report so that a reader can judge them; both were published for regression and machine-learning models alike. A field-wide appraisal of psychiatric prediction models found calibration reported in 22 % and adequate internal validation in 11 % [15]. What is missing is the link between the two literatures: an appraisal tells us how common a practice is, but not how much it changes the numbers; a re-analysis on one dataset gives a number, but not a sense of how widespread the problem is. This paper supplies the link on the cohorts where the problem is most visible.

1.3 Gaps and contributions

No study has (i) appraised the published models on these cohorts systematically, (ii) measured the paired difference between record-wise and subject-wise evaluation on the same data, models and seeds, (iii) reported cross-cohort transfer with the shared controls handled, or (iv) reported calibration for any model on these cohorts. This paper does all four in one design (Figure 1) and contributes:

1. a closed-scope systematic appraisal (Part I) with PROBAST+AI and TRIPOD+AI, whose outcomes are chosen so that each can be priced on the data (Section 2);
2. a controlled re-evaluation (Part II) that quantifies leakage inflation (E1), cross-cohort loss with calibration collapse (E2), the calibration blind spot with a recalibration test (E3), sex-stratified performance (E4), transdiagnostic and five-class tasks on the de-duplicated 162-participant release (E5), and an ablation over feature groups, resampling and cross-validation scheme (E6), with participant-level bootstrap uncertainty throughout;
3. a comparison of every published claim against the measured honest and leaky bands (Table 13, Figure 2), a calibration comparison (Figure 5) and a multi-metric profile of the models (Figure 6);
4. a public, tested codebase in which subject-wise splitting is enforced by a build-failing test, cross-cohort overlap is detected from the data, every run writes a manifest, and every table in this paper is generated from those manifests, together with a TRIPOD+AI self-audit.

The paper is written for readers who build and evaluate ML models; the clinical background is kept to what is needed to interpret the results, and every evaluation concept is defined where it is first used. Section 2 reports the appraisal; Section 3 reports the re-evaluation; Section 4 discusses mechanism, implications and limitations.

2 Part I: systematic appraisal of the published models

2.1 Design

Part I is a closed-scope systematic appraisal reported following PRISMA 2020 principles; it was not prospectively registered, and the Part II analysis plan is fixed in versioned configuration files in the repository. *Eligibility*: peer-reviewed articles, indexed conference papers and preprints (January 2018 to September 2026, English) that train or evaluate a supervised model to predict group membership, symptom score or state from DEPRESJON, PSYKOSE, HYPERAKTIV or OBF-Psychiatric activity data and report at least one performance metric; excluded are reviews, dataset descriptors without a model, and unsupervised or purely descriptive analyses. *Identification*: forward citation tracing of the four dataset papers and searches for the dataset names (“Depresjon”, “Psykoese”, “Hyperaktiv”, “OBF-Psychiatric”) in Google Scholar, Semantic Scholar, PubMed/PMC, arXiv and the publishers’ sites, run in September 2026 (Supplementary Table S3 records sources and dates). *Screening and extraction*: titles, abstracts and, where accessible, full texts were screened and extracted by the authors against a fixed codebook; extractions that could rest only on the abstract and metadata are marked as such in Table 1 and treated as a limitation. *Appraisal*: PROBAST+AI (four domains) and the 27-item TRIPOD+AI checklist in duplicate. *Pre-specified outcomes per study*: unit of splitting (participant, day/window, unstated); any external validation; calibration reported; AUROC reported or accuracy only; participant-level events per candidate predictor; use of both DEPRESJON and PSYKOSE and, if so, recognition of the shared control group; code availability; and the best reported performance, so that claims can be plotted against the Part II bands.

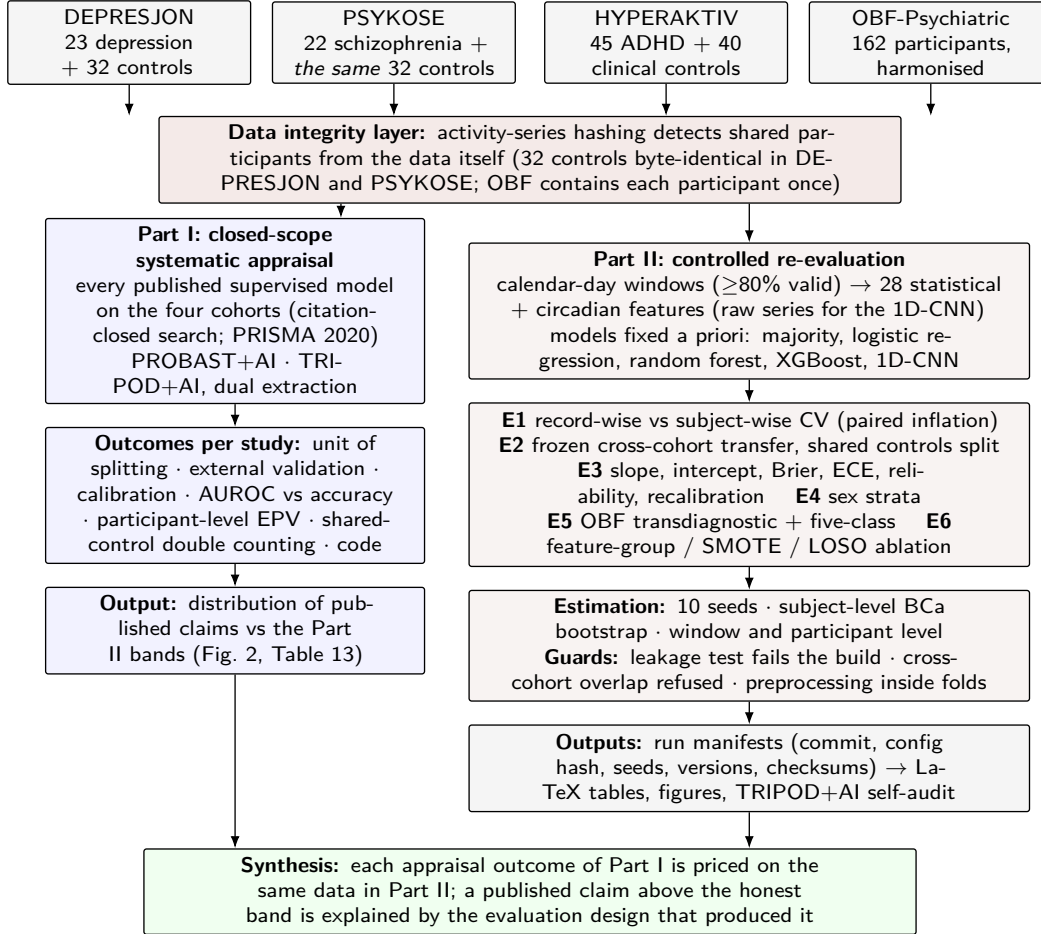


Figure 1: Architecture of the study. Four open actigraphy cohorts pass through a data-level integrity check (participant hashing) into a closed-scope systematic appraisal of the published models (Part I) and a controlled re-evaluation under the designs the appraisal scores (Part II); the synthesis maps every appraisal outcome to a measured cost.

2.2 Results of Part I

Twenty-eight eligible studies were identified (Table 1; sources in Supplementary Table S4): 13 on DEPRESJON alone, 3 on PSYKOSE alone, 6 on HYPERAKTIV, 2 on OBF-Psychiatric and 4 combining DEPRESJON with PSYKOSE. Seven split by participant (the four dataset papers and three follow-ups), 9 split by day or window, and 12 did not state the unit in the available text. None reported any calibration measure. None validated a model on a second cohort; one preprint claims an independent set under a “modified leave-one-out” scheme. Four released code, all of them dataset papers. Every study reporting accuracy above 0.85 on DEPRESJON or above 0.90 on PSYKOSE used, or did not state, a day-level unit (Figure 2). Of the four studies that combined DEPRESJON and PSYKOSE, the OBF-Psychiatric descriptor lists the shared controls once by construction; none of the other three states how the shared control group was handled. Participant-level events per candidate predictor cannot exceed 23 cases divided by the number of candidate predictors on DEPRESJON and 22 on PSYKOSE, i.e. below 1 for any feature set larger than the number of cases, and no study reported the quantity.

Table 1: Part I: eligible published supervised models on the four cohorts (sources in Supplementary Table S4). “Extracted from the abstract and metadata (full text not accessible at extraction); n.r. = not reported there. Split unit: participant = leave-one-participant-out or grouped folds; day/window = folds over days or windows; unstated = not stated in the available text.

ID	First author	Year	Cohorts	Best model	Split unit	Best reported	Calib.	Ext. val.	Code
S01	Garcia-Coja	2018	DEPRESJON	Random forest / DNN	participant (leave-one-patient-out)	F1 0.73	no	no	yes
S02*	Zanella-Calzada	2019	DEPRESJON	Random forest	day / window	accuracy 0.89	no	no	no
S03*	Froger	2019	DEPRESJON	1D-CNN	participant (leave-one-participant-out)	F1 n.r.	no	no	no
S04*	Jakobsen	2020	DEPRESJON	Random forest / DNN / CNN	participant (leave-one-out)	accuracy n.r.	no	no	no
S05	Jakobsen	2020	PSYKOSE	Classical ML	participant (leave-one-patient-out)	F1 n.r.	no	no	yes
S06	Hicks	2021	HYPERAKTIV	Classical ML	unstated (10-fold on features)	accuracy n.r.	no	no	yes
S07*	Rodriguez-Ruiz	2022	DEPRESJON	2D-CNN	day / window	accuracy 0.767	no	no	no
S08*	Rodriguez-Ruiz	2022	DEPRESJON+PSYKOSE	CNN	day / window	accuracy n.r.	no	no	no
S09*	Kumar	2022	DEPRESJON	ML on autoregressive features	unstated	accuracy 0.78	no	no	no
S10*	Ahmed (arXiv)	2023	DEPRESJON	Transfer learning ensemble	modified leave-one-out	accuracy 0.96	no	claims independent set	no
S11*	Hybrid RF-ANN (arXiv)	2023	DEPRESJON	RF-ANN ensemble	unstated	accuracy n.r.	no	no	no
S12*	UMAP study	2023	DEPRESJON	ML + UMAP	unstated	accuracy n.r.	no	no	no
S13*	Genetic-algorithm feature selection	2023	DEPRESJON	GA feature selection + classifier	day / window (intervals)	accuracy n.r.	no	no	no
S14*	Attention CNN (IoMT)	2023	PSYKOSE	Deep CNN + attention	day / window	accuracy n.r.	no	no	no
S15*	Multi-branch DL	2024	PSYKOSE	Multi-branch deep network	day / window	accuracy 0.92	no	no	no
S16*	DL + XAI	2024	PSYKOSE (+DEPRESJON)	Deep network + XAI	day / window (24-h segments)	accuracy 0.94	no	no	no
S17*	Hopping-mean augmentation	2024	DEPRESJON	Augmented ML	day / window	accuracy n.r.	no	no	no
S18*	IoMT dimensionality reduction	2024	HYPERAKTIV	10 classifiers on reduced features	unstated	accuracy n.r.	no	no	no
S19*	Explainable XGBoost	2025	DEPRESJON	XGBoost + ADASYN	unstated in abstract	accuracy 0.849	no	no	no
S20*	Improved CNN framework	2025	DEPRESJON+PSYKOSE	Improved CNN	day / window	accuracy n.r.	no	no	no
S21*	Enhanced ADHD ML	2025	HYPERAKTIV/OBF (45 participants)	SVM	unstated	F1 0.779	no	no	no
S22*	HYPERAKTIV ML study	2025	HYPERAKTIV	RF / LightGBM / SVM	unstated	accuracy n.r.	no	no	no
S23*	Multimodal digital biomarkers	2025	HYPERAKTIV	Bernoulli NB (actigraphy+HRV)	unstated	F1 0.79	no	no	no
S24	OBF-Psychiatric descriptor	2025	OBF-Psychiatric	Classical ML baselines	participant	accuracy n.r.	no	no	yes
S25*	Data-uncertainty case study	2025	OBF-Psychiatric (ADHD)	QC + ML	participant	nan no difference after QC	no	no	no
S26*	ViT / CoAtNet on images	2025	DEPRESJON+PSYKOSE	CoAtNet-Tiny	participant (three-fold subject-wise)	accuracy n.r.	no	no	no
S27*	Time-segmentation evaluation	2026	DEPRESJON	ML on day segments	n.r.	accuracy n.r.	no	no	no
S28*	Accurate ADHD identification	2022	HYPERAKTIV	ML on real-time activity	unstated	accuracy n.r.	no	no	no

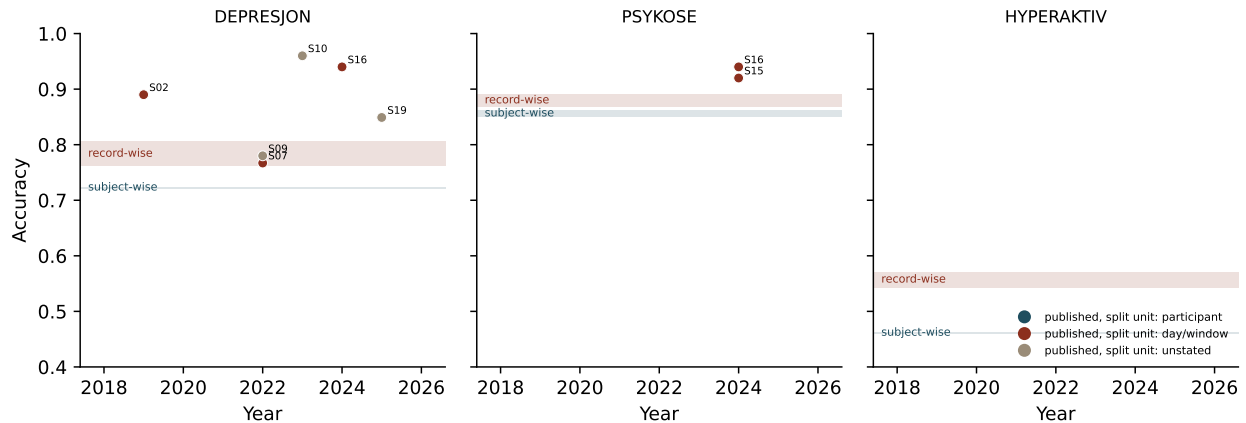


Figure 2: Published best accuracies (points, labelled by study ID in Table 1, coloured by the unit of splitting) against the window-level accuracy bands measured in Part II under subject-wise and record-wise five-fold cross-validation (shaded; range over logistic regression, random forest and XGBoost). The vertical axis is classification accuracy for both. Studies that report a metric other than accuracy, or no numeric value in the available text, are not plotted; all are listed in Table 1.

2.3 Conclusion of Part I

The published literature on these cohorts reports discrimination that, where it exceeds the dataset authors’ baselines, coincides with day-level or unstated splitting (21 of 28 studies, 75%, Wilson 95% CI 57 to 87%); it does not report calibration (0 of 28, 0 to 12%); it does not test transfer between cohorts (0 of 28); and it rarely releases code (4 of 28, 14%, 6 to 31%). Under PROBAST+AI these are high-risk judgements in the analysis domain, through the signalling questions on optimism and data leakage, on evaluation of calibration as well as discrimination, and on sample size relative to the number of candidate predictors; under TRIPOD+AI they correspond to absent reporting of the internal-validation design, of calibration performance, of the sample-size rationale and of code availability. The appraisal is descriptive by construction: it establishes which practices are prevalent, not how much they matter. Part II measures that on the same data.

3 Part II: controlled re-evaluation

3.1 Rationale and design principles

Part II is designed so that each of the appraisal outcomes in Part I corresponds to one controlled contrast on the same data, and so that the contrast cannot be attributed to anything other than the practice under study. Four principles follow. First, *models are fixed a priori*: the same four estimators with the same hyper-parameters are used in every experiment, so that a difference between two arms is a difference in evaluation design rather than in model selection. Second, *the unit of inference is the participant*: every split, every bootstrap resample and every aggregate is keyed on participant identity, and window-level numbers are reported only because they are the numbers the literature reports. Third, *integrity is enforced by code rather than by discipline*: a test fails the build if any experiment other than E1 or E5 requests record-wise splitting, the cross-validation runner asserts fold disjointness at run time, the external-validation runner refuses any cohort pair with a participant on both sides, and all preprocessing lives inside a pipeline fitted on the training fold. Fourth, *every number is traceable*: each run writes a manifest with the git commit, configuration hash, seeds, package versions and dataset checksums, and every table and figure in this paper is written by the code from those runs.

3.2 Data, participants and windows

Table 2: Participant characteristics per cohort. HYPERAKTIV controls are psychiatric patients without ADHD (clinical controls); DEPRESJON and PSYKOSE share the same 32 healthy controls. Sex, age band and recording days from the harmonised OBF-Psychiatric metadata; day-windows are the analysis units after the 80% validity threshold.

Cohort	Group	n	Female, n (%)	Age (modal band)	Recording days, median (range)	Valid day-windows	Clinical scale, median (IQR)
depresjon	cases (depression)	23	10 (43)	45-49	13 (5-18)	405	MADRS 24 (IQR 18-26)
depresjon	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	–
psykose	cases (schizophrenia)	22	3 (14)	35-39	13 (12-14)	390	BPRS 51 (IQR 44-58)
psykose	controls (healthy_control)	32	20 (62)	50-54	13 (8-20)	739	–
hyperaktiv	cases (adhd)	45	21 (47)	30-39	7 (2-10)	343	ASRS 49 (IQR 40-55)
hyperaktiv	controls (clinical_control)	40	20 (50)	17-29	7 (6-10)	329	ASRS 34 (IQR 25-43)
obf_psychiatric	cases (adhd, clinical, depression, schizophrenia)	130	54 (42)	30-39	7 (2-18)	1467	–
obf_psychiatric	controls (control)	32	20 (62)	50-54	13 (8-20)	739	–

DEPRESJON [1] comprises 23 patients with unipolar or bipolar depression (MADRS at the start of recording, median 24, interquartile range 18 to 26) and 32 healthy controls; PSYKOSE [2] comprises 22 inpatients with schizophrenia (BPRS median 51, 44 to 58) and the same 32 controls; HYPERAKTIV [3] comprises adults referred for ADHD assessment, of whom 45 with ADHD and 40 clinical controls (psychiatric outpatients without ADHD) have activity recordings, read here from the OBF-Psychiatric release [4]. All cohorts were recorded with an Actiwatch (Cambridge Neurotechnology) at one-minute resolution; recruitment, consent and clinical assessment

are described in the source publications. Table 2 summarises the analysed participants; sex, age band and recording days are taken from the harmonised OBF metadata and matched to the original cohorts by activity-series hash.

Hashing each participant’s activity series confirmed two facts that the documentation states but the literature rarely acts on: the 32 PSYKOSE control recordings are byte-identical to the 32 DEPRESJON control recordings, and OBF-Psychiatric contains each of the 162 participants exactly once ($23 + 22 + 85 + 32$). The analysis unit was one calendar day (00:00 to 23:59) with at least 80% valid minutes; partial first and last days were dropped. DEPRESJON yielded 1,034 valid days (median 15 per participant), PSYKOSE 1,021 (16) and HYPERAKTIV 502 (6), 1,818 unique days in total; no participant lost all days. Labels are participant-level diagnostic groups inherited by every window, and the data schema rejects a participant carrying two labels.

3.3 Predictors

Twenty-eight predictors were computed per day (Supplementary Table S2), chosen to cover the constructs the actigraphy literature associates with mood and psychotic disorders rather than to maximise discrimination: distributional statistics of the minute counts (mean, SD, coefficient of variation, median, 10th and 90th percentiles, maximum, skewness, kurtosis, proportion of zero-count minutes, mean and SD of $\log(1 + x)$); day/night contrasts (means for 08:00 to 22:00 and for the night, their ratio, night-time inactivity); non-parametric circadian measures on the hourly profile (M10, L5, relative amplitude, intradaily variability, sine and cosine of the M10-onset hour); autocorrelation at 5 and 60 minutes; spectral summaries (mean log power, dominant within-day period); the number of rest/activity transitions; and the missing-minute fraction. No demographic variable was used, so that the cross-cohort comparison in E2 is a comparison of populations and not of covariate availability. A raw representation, the $\log(1 + x)$ -transformed 1440-minute series, feeds a one-dimensional convolutional network in a supplementary arm.

3.4 Models and preprocessing

Four estimators were fixed before any result was seen: a majority-class baseline (constant prediction at the training prevalence); L_2 -regularised logistic regression ($C = 1$, class-balanced weights); a random forest (500 trees, minimum leaf size 2, balanced subsampling); and XGBoost (300 rounds, maximum depth 3, learning rate 0.05, row and column subsampling 0.8). The two simplest exist so that the flexible ones must earn their complexity. Hyper-parameters are library defaults or conservative values and were not tuned, because tuning inside subject-wise folds would raise honest performance somewhat without altering the paired inflation that E1 estimates, while tuning inside record-wise folds would inflate it further. Median imputation and standardisation are fitted on the training fold only, inside an `imblearn` pipeline whose sampler steps run only at training time; no resampling was used in the main runs (E6 tests SMOTE inside folds). The supplementary 1D-CNN has three convolution–pooling blocks, global average pooling and a linear head, trained for a fixed 30 epochs with a class-weighted loss; it has no early stopping on the test fold.

3.5 Performance measures and estimation

Discrimination is reported as AUROC (primary), accuracy, balanced accuracy, macro-F1 and Matthews correlation at a 0.5 threshold, and area under the precision–recall curve. Calibration is reported as the Brier score, the calibration slope (coefficient of the logit of the predicted probability in a logistic regression of the outcome) and the calibration intercept (fitted with the slope fixed at one) [12], the expected calibration error over ten equal-width bins, and reliability curves. All measures are computed at two levels: the *window* level, on every day’s prediction, and the *participant* level, on the mean predicted probability per participant, which is the quantity a screening decision would use.

Every experiment is repeated over ten seeds that govern fold assignment and model randomness. Point estimates are computed on the seed-averaged out-of-fold predictions (an ensemble over seeds; the per-seed mean and SD are also reported), and 95% intervals are bias-corrected and accelerated bootstrap intervals from 1,000 resamples of *participants*, drawing all of a participant’s windows together and using a leave-one-participant-out jackknife for the acceleration term. Resampling windows instead of participants would treat correlated observations as independent and produce intervals that are far too narrow. Participant-level events per predictor (EPV) are reported as the minority-class participants divided by 28.

3.6 Experiments

E1: the cost of leaky splitting. Five-fold cross-validation was run under two splitters on identical data and seeds. The *subject-wise* splitter is a stratified group k -fold in which no participant appears on both sides of any fold. The *record-wise* splitter is a stratified k -fold over days that ignores participant identity, so that days from one participant appear in training and test. The estimand is the paired difference in AUROC (record-wise minus subject-wise) for the same participants, bootstrapped by resampling participants in both arms simultaneously. This is the design most day-level papers on these cohorts use; the experiment measures its optimism directly rather than inferring it.

E2: the cost of skipping external validation. For each ordered cohort pair, the model was fitted on all windows of cohort A and applied, frozen, to cohort B; the internal comparator is subject-wise cross-validation on A. Because DEPRESJON and PSYKOSE share their controls, each shared control was assigned at random (seed 0) to exactly one cohort, 16 to each, so that both cohorts retain both classes and no participant crosses; the assignment table is in the repository. Two pairs are informative (DEPRESJON $\leftrightarrow$ PSYKOSE: same controls, different patient group, same device); the third, DEPRESJON $\rightarrow$ HYPERAKTIV, is a deliberate label-shift control in which a depression-versus-healthy model is applied to ADHD-versus-clinical-control data and is expected to fail.

E3: the calibration blind spot. For every model in E1 (subject-wise arm) and E2, the calibration measures above were computed, and a logistic recalibration (slope and intercept) fitted on the training cohort’s out-of-fold predictions was applied to the external predictions, to test whether external miscalibration is repairable without access to the target cohort.

E4: sex strata. Subject-wise E1 predictions were stratified by sex and the measures recomputed with participant-level intervals; strata with fewer than six participants or one class are omitted.

E5: OBF-Psychiatric. On the harmonised release, with its single de-duplicated control group, two tasks were evaluated under both splitters: any psychiatric group versus healthy control (130 versus 32 participants), and the five diagnostic groups (macro one-versus-rest AUROC).

E6: ablation. Under subject-wise cross-validation, with logistic regression and XGBoost and five seeds: each feature group dropped and each used alone (distributional, day/night, circadian, temporal/spectral); SMOTE applied inside training folds; and leave-one-subject-out in place of five-fold. E6 answers the objection that honest performance is low only because the feature set, the class balance or the fold count was unfavourable.

Table 3: E1, window-level AUROC under subject-wise versus record-wise five-fold cross-validation. Estimates on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	AUROC subject-wise	AUROC record-wise	Inflation (paired)
depresjon	Majority class	0.465 [0.292, 0.653]	0.491 [0.459, 0.528]	0.026 [-0.147, 0.218]
depresjon	Logistic regression	0.779 [0.675, 0.861]	0.845 [0.781, 0.896]	0.065 [0.024, 0.122]
depresjon	Random forest	0.770 [0.681, 0.848]	0.877 [0.823, 0.923]	0.107 [0.058, 0.168]
depresjon	XGBoost	0.764 [0.675, 0.842]	0.878 [0.830, 0.918]	0.114 [0.060, 0.181]
psykose	Majority class	0.437 [0.267, 0.632]	0.500 [0.470, 0.535]	0.062 [-0.113, 0.241]
psykose	Logistic regression	0.907 [0.820, 0.942]	0.927 [0.875, 0.955]	0.020 [0.006, 0.041]
psykose	Random forest	0.893 [0.798, 0.934]	0.937 [0.890, 0.962]	0.044 [0.017, 0.080]
psykose	XGBoost	0.903 [0.810, 0.941]	0.945 [0.907, 0.969]	0.042 [0.016, 0.078]
hyperaktiv	Majority class	0.493 [0.355, 0.628]	0.483 [0.433, 0.533]	-0.010 [-0.170, 0.148]
hyperaktiv	Logistic regression	0.449 [0.360, 0.535]	0.579 [0.486, 0.661]	0.130 [0.095, 0.167]
hyperaktiv	Random forest	0.443 [0.361, 0.534]	0.612 [0.526, 0.692]	0.169 [0.135, 0.206]
hyperaktiv	XGBoost	0.442 [0.360, 0.524]	0.593 [0.518, 0.664]	0.151 [0.116, 0.186]

Table 4: E1, participant-level AUROC under subject-wise versus record-wise five-fold cross-validation. Estimates on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	AUROC subject-wise	AUROC record-wise	Inflation (paired)
depresjon	Majority class	0.423 [0.300, 0.551]	0.427 [0.295, 0.555]	0.001 [-0.191, 0.202]
depresjon	Logistic regression	0.899 [0.822, 0.969]	0.965 [0.927, 0.990]	0.065 [0.011, 0.109]
depresjon	Random forest	0.883 [0.812, 0.948]	0.977 [0.949, 1.000]	0.094 [0.041, 0.146]
depresjon	XGBoost	0.874 [0.795, 0.947]	0.985 [0.964, 1.000]	0.111 [0.040, 0.175]
psykose	Majority class	0.419 [0.278, 0.557]	0.462 [0.331, 0.593]	0.043 [-0.148, 0.227]
psykose	Logistic regression	0.974 [0.943, 1.000]	0.994 [0.983, 1.000]	0.020 [0.000, 0.043]
psykose	Random forest	0.956 [0.913, 0.997]	0.994 [0.983, 1.000]	0.038 [0.000, 0.074]
psykose	XGBoost	0.969 [0.935, 1.000]	0.997 [0.991, 1.000]	0.028 [0.000, 0.059]
hyperaktiv	Majority class	0.497 [0.398, 0.597]	0.443 [0.349, 0.540]	-0.054 [-0.203, 0.106]
hyperaktiv	Logistic regression	0.424 [0.328, 0.519]	0.601 [0.502, 0.695]	0.177 [0.131, 0.221]
hyperaktiv	Random forest	0.424 [0.333, 0.518]	0.641 [0.550, 0.722]	0.217 [0.177, 0.257]
hyperaktiv	XGBoost	0.438 [0.337, 0.530]	0.652 [0.555, 0.732]	0.214 [0.173, 0.261]

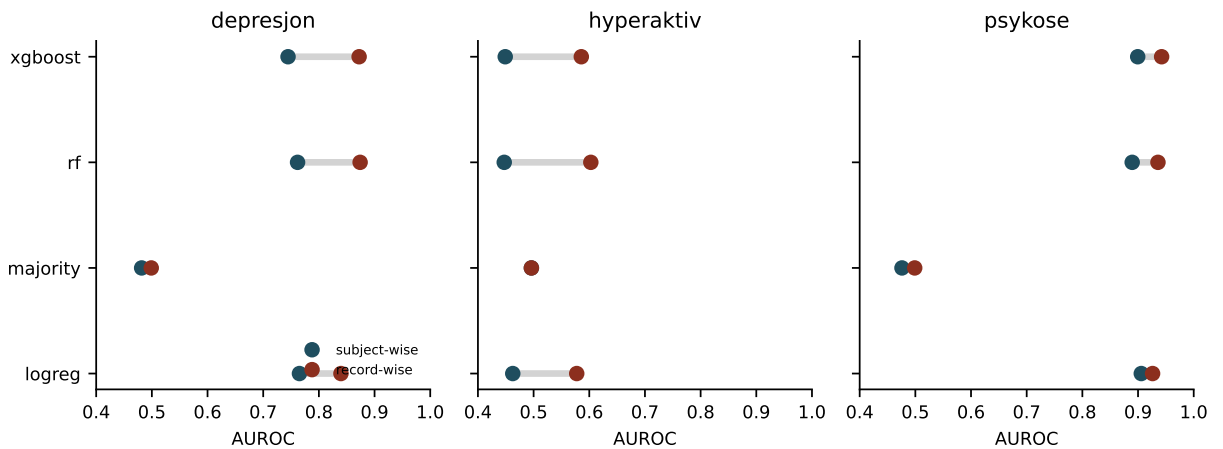


Figure 3: E1. Window-level AUROC of each model under subject-wise (dark) and record-wise (red) five-fold cross-validation; the grey bar is the leakage inflation.

3.7 E1 results: the cost of leaky splitting

Record-wise splitting raised AUROC for every informative model in every cohort, at both levels (Tables 3 and 4, Figure 3). On DEPRESJON, XGBoost rose from 0.764 (95% CI 0.675 to 0.842) to 0.878 (0.830 to 0.918) at window level, a paired inflation of 0.114 (0.060 to 0.181); at participant level the honest 0.874 became 0.985, and random forest and logistic regression showed the same pattern with inflations of 0.107 and 0.065. On PSYKOSE, where schizophrenia produces a large and consistent activity deficit and honest discrimination is already 0.893 to 0.907, the inflation was smaller (0.020 to 0.044) but still excluded zero, and the participant-level leaky estimates were 0.994 to 0.997. On HYPERAKTIV the honest estimate was at or below chance for every model (0.442 to 0.449 at window level, 0.424 to 0.438 at participant level), consistent with a recent report that quality-controlled OBF activity features do not separate ADHD from controls [4]; record-wise splitting nevertheless produced 0.579 to 0.612 at window level and 0.601 to 0.652 at participant level, an inflation of 0.130 to 0.169 with intervals excluding zero. Leakage did not merely exaggerate a real signal; it created one. Across cohorts the ordering was stable: logistic regression inflated least (0.065, 0.020, 0.130) and the tree ensembles most (up to 0.114, 0.044, 0.169), as expected if the mechanism is memorisation of participant-specific activity signatures. Accuracy and macro-F1 behaved identically (Table 5).

The 1D-CNN on the raw series did not earn its complexity. Under subject-wise splitting its seed-averaged AUROC was 0.696 (0.576 to 0.795) on DEPRESJON, 0.849 (0.744 to 0.910) on PSYKOSE and 0.459 (0.356 to 0.588) on HYPERAKTIV, below the feature-based models in every cohort, and it was unstable across seeds (per-seed means 0.61, 0.74 and 0.48). Its window-level inflation under record-wise splitting was small, but its participant-level estimate rose from 0.738 to 0.912 on DEPRESJON and from 0.922 to 0.986 on PSYKOSE (Tables 6 and 7): the network learned to recognise participants across days.

Table 5: E1: window-level discrimination and calibration for every model, cohort and splitting design (per-seed means over ten seeds).

Cohort	Model	Split	Accuracy	Macro-F1	AUROC	Brier	Cal. slope	Cal. intercept	ECE
depresjon	Majority class	subject-wise	0.653	0.395	0.482	0.227	-4.01	-0.00	0.000
depresjon	Majority class	record-wise	0.653	0.395	0.499	0.227	0.29	0.00	0.000
depresjon	Logistic regression	subject-wise	0.722	0.706	0.765	0.194	0.52	-0.54	0.118
depresjon	Logistic regression	record-wise	0.761	0.750	0.840	0.163	0.90	-0.62	0.095
depresjon	Random forest	subject-wise	0.724	0.679	0.762	0.185	0.69	0.01	0.061
depresjon	Random forest	record-wise	0.806	0.782	0.874	0.136	1.23	-0.11	0.039
depresjon	XGBoost	subject-wise	0.722	0.676	0.745	0.202	0.43	0.29	0.123
depresjon	XGBoost	record-wise	0.802	0.776	0.872	0.137	0.82	0.07	0.039
psykose	Majority class	subject-wise	0.661	0.398	0.476	0.224	-3.95	0.00	0.000
psykose	Majority class	record-wise	0.661	0.398	0.499	0.224	0.31	0.00	0.000
psykose	Logistic regression	subject-wise	0.856	0.842	0.906	0.115	0.90	-0.60	0.086
psykose	Logistic regression	record-wise	0.868	0.855	0.926	0.103	1.07	-0.62	0.078
psykose	Random forest	subject-wise	0.850	0.829	0.889	0.118	0.95	-0.12	0.048
psykose	Random forest	record-wise	0.878	0.861	0.936	0.093	1.32	-0.17	0.048
psykose	XGBoost	subject-wise	0.862	0.842	0.899	0.108	0.71	0.15	0.046
psykose	XGBoost	record-wise	0.891	0.876	0.942	0.082	0.91	0.07	0.026
hyperaktiv	Majority class	subject-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Majority class	record-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Logistic regression	subject-wise	0.461	0.460	0.463	0.286	-0.21	-0.01	0.160
hyperaktiv	Logistic regression	record-wise	0.543	0.543	0.577	0.252	0.21	0.01	0.074
hyperaktiv	Random forest	subject-wise	0.463	0.462	0.447	0.286	-0.32	0.00	0.156
hyperaktiv	Random forest	record-wise	0.571	0.571	0.603	0.245	0.57	0.00	0.055
hyperaktiv	XGBoost	subject-wise	0.460	0.460	0.449	0.330	-0.15	0.00	0.248
hyperaktiv	XGBoost	record-wise	0.557	0.557	0.586	0.271	0.26	0.00	0.151

3.8 E2 results: the cost of skipping external validation

Transfer was asymmetric (Table 8). Models trained on DEPRESJON, whose positive class is the milder activity deficit, separated schizophrenia from the held-out controls as well as they had

Table 6: E1, window-level AUROC under subject-wise versus record-wise five-fold cross-validation. Estimates on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	AUROC subject-wise	AUROC record-wise	Inflation (paired)
depresjon	1D-CNN	0.696 [0.576, 0.795]	0.729 [0.671, 0.784]	0.033 [-0.063, 0.122]
psykose	1D-CNN	0.849 [0.744, 0.910]	0.892 [0.836, 0.935]	0.043 [-0.000, 0.095]
hyperaktiv	1D-CNN	0.459 [0.356, 0.588]	0.513 [0.447, 0.578]	0.054 [-0.079, 0.189]

Table 7: E1, participant-level AUROC under subject-wise versus record-wise five-fold cross-validation. Estimates on seed-averaged out-of-fold predictions with subject-level BCa 95% bootstrap intervals; inflation is the paired record-wise minus subject-wise difference.

Cohort	Model	AUROC subject-wise	AUROC record-wise	Inflation (paired)
depresjon	1D-CNN	0.738 [0.634, 0.844]	0.912 [0.837, 0.959]	0.174 [0.067, 0.270]
psykose	1D-CNN	0.922 [0.852, 0.976]	0.986 [0.963, 1.000]	0.064 [0.016, 0.111]
hyperaktiv	1D-CNN	0.455 [0.371, 0.563]	0.529 [0.432, 0.622]	0.074 [-0.060, 0.208]

Table 8: E2: internal (subject-wise CV on the training cohort) versus external (unchanged model applied to the second cohort) performance. Window-level estimates, mean over seeds, subject-level BCa 95% intervals. EPV = minority-class subjects per candidate predictor in the training cohort.

Train → Test	Model	EPV	Internal AUROC	External AUROC	Δ	Int. slope	Ext. slope	Ext. intercept	Ext. Brier
depresjon → psykose	Majority class	0.6	0.465 [0.224, 0.662]	0.500 [0.500, 0.500]	+0.035	-4.16	0.00	-0.02	0.250
depresjon → psykose	Logistic regression	0.6	0.779 [0.701, 0.871]	0.779 [0.676, 0.847]	-0.000	0.66	0.50	0.06	0.193
depresjon → psykose	Random forest	0.6	0.770 [0.607, 0.837]	0.776 [0.688, 0.846]	+0.006	0.58	0.80	-0.02	0.193
depresjon → psykose	XGBoost	0.6	0.764 [0.608, 0.825]	0.769 [0.678, 0.835]	+0.005	0.34	0.55	-0.01	0.204
psykose → depresjon	Majority class	0.6	0.390 [0.206, 0.601]	0.500 [0.500, 0.500]	+0.110	-4.07	0.00	0.06	0.250
psykose → depresjon	Logistic regression	0.6	0.885 [0.790, 0.938]	0.689 [0.564, 0.793]	-0.196	0.72	0.27	1.20	0.262
psykose → depresjon	Random forest	0.6	0.854 [0.756, 0.926]	0.728 [0.620, 0.818]	-0.126	0.76	0.63	0.74	0.225
psykose → depresjon	XGBoost	0.6	0.877 [0.793, 0.936]	0.671 [0.569, 0.770]	-0.206	0.55	0.30	1.44	0.285
depresjon → hyperaktiv	Majority class	0.8	0.465 [0.292, 0.653]	0.500 [0.500, 0.500]	+0.035	-4.01	-0.01	0.65	0.275
depresjon → hyperaktiv	Logistic regression	0.8	0.779 [0.675, 0.861]	0.442 [0.348, 0.532]	-0.338	0.52	-0.16	-0.42	0.350
depresjon → hyperaktiv	Random forest	0.8	0.770 [0.681, 0.848]	0.461 [0.360, 0.557]	-0.310	0.69	-0.15	0.33	0.317
depresjon → hyperaktiv	XGBoost	0.8	0.764 [0.675, 0.842]	0.459 [0.358, 0.550]	-0.305	0.43	-0.09	0.48	0.364

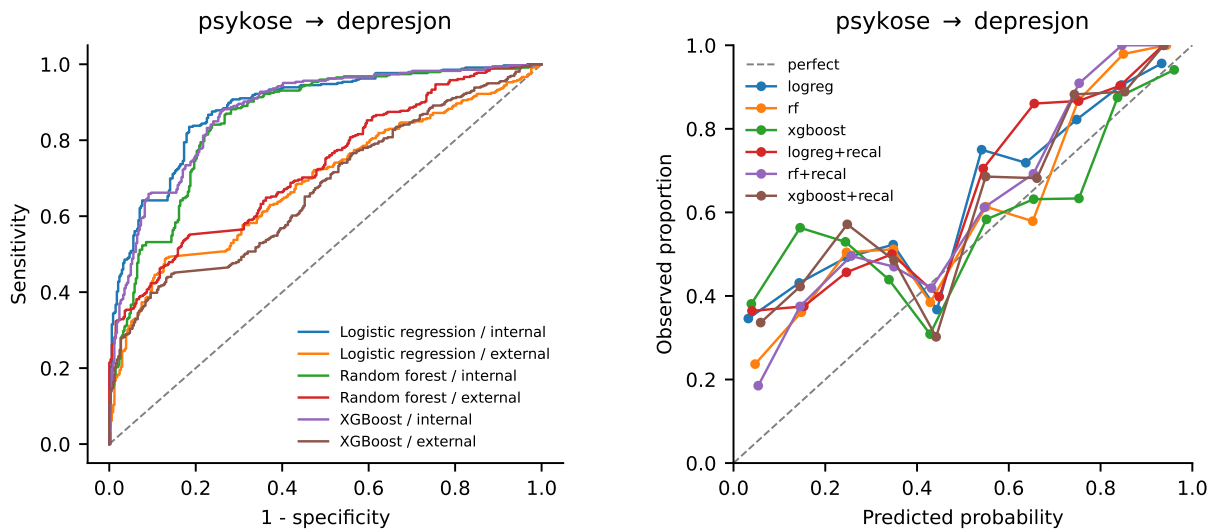


Figure 4: E2/E3, PSYKOSE → DEPRESJON. Left: ROC curves of the internal (subject-wise CV on PSYKOSE) and external (frozen model on DEPRESJON) predictions. Right: reliability curves of the external predictions before and after logistic recalibration fitted on the training cohort.

separated depression internally: external AUROC 0.769 to 0.779 against internal 0.764 to 0.779, with calibration slopes of 0.50 to 0.81 and intercepts near zero. Models trained on PSYKOSE, whose positive class is the more severe deficit, lost 0.13 to 0.21 AUROC when applied to DEPRESJON (logistic regression 0.885 $\rightarrow$ 0.689; random forest 0.854 $\rightarrow$ 0.728; XGBoost 0.877 $\rightarrow$ 0.671), and their probabilities were both overconfident and shifted: slopes 0.27 to 0.63, intercepts 0.74 to 1.44, Brier scores 0.22 to 0.29 against 0.25 for a constant prediction (Figure 4). A model that had looked excellent internally would have been worse than useless as a probability on the new cohort. The label-shift pair DEPRESJON $\rightarrow$ HYPERAKTIV fell to AUROC 0.442 to 0.460 with negative slopes, as designed; a paper that reported this transfer as “external validation” of a mental-disorder detector would be reporting noise. Participant-level EPV in the training cohorts was 0.6 (DEPRESJON and PSYKOSE after the control split) and 0.8 (HYPERAKTIV), one to two orders of magnitude below any guidance, while the window-level counts (hundreds of days per class) would have suggested adequacy.

3.9 E3 results: the calibration blind spot

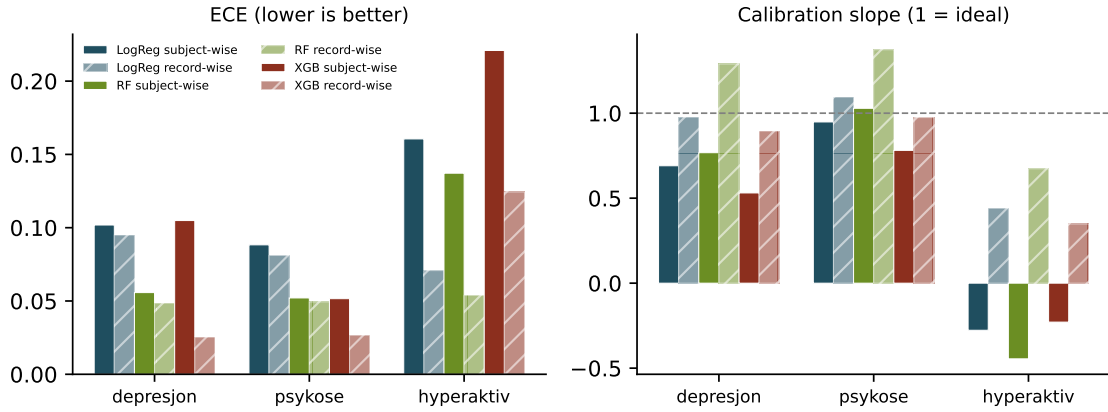


Figure 5: E3. Expected calibration error (left) and calibration slope (right) of every model under subject-wise (solid) and record-wise (hatched) cross-validation. Record-wise models appear better calibrated because they score participants they have seen.

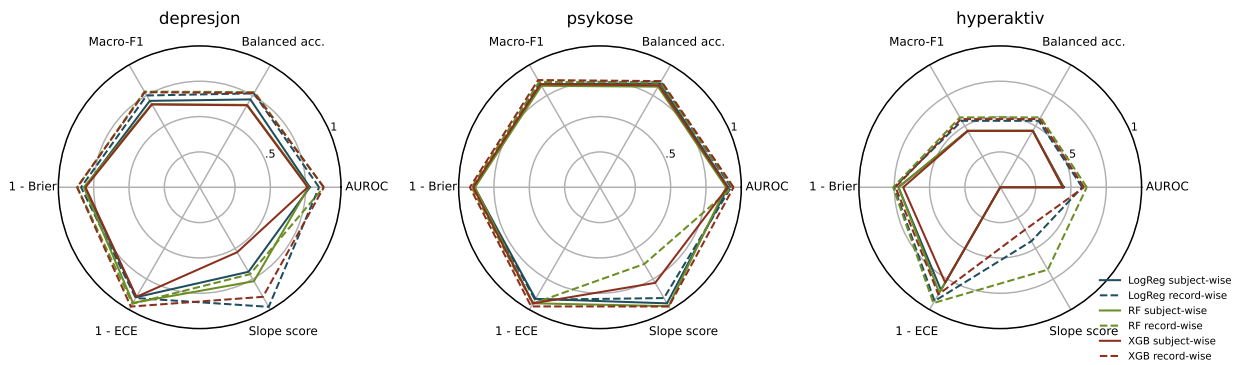


Figure 6: Multi-metric profile of the models per cohort (AUROC, balanced accuracy, macro-F1, 1–Brier, 1–ECE and a slope score $1 - |1 - \text{slope}|$) under subject-wise (solid) and record-wise (dashed) cross-validation. Leakage expands every axis at once; on HYPERAKTIV the honest profile collapses to chance while the leaky profile does not.

Under honest splitting the informative models were already miscalibrated (Table 9, Figures 5 and 6). On DEPRESJON, slopes were 0.53 (XGBoost), 0.69 (logistic regression) and 0.77 (random forest), with ECE 0.06 to 0.11; on PSYKOSE, 0.95 for logistic regression; on HYPERAKTIV,

slopes were negative and ECE 0.14 to 0.22, i.e. the models had learned nothing and yet issued confident probabilities. Under record-wise splitting the same models appeared markedly better calibrated (slopes 0.9 to 1.3, ECE 0.02 to 0.05 on DEPRESJON and PSYKOSE), which is not a virtue but a symptom: they were scoring participants they had seen, and their confidence was justified only for those participants. The radar profiles make the same point in one view: leakage expands every axis simultaneously, and on HYPERAKTIV the honest profile collapses to the centre while the leaky profile does not. Recalibrating external predictions with a slope and intercept fitted on the training cohort's own out-of-fold predictions repaired the easy direction (DEPRESJON $\rightarrow$ PSYKOSE: slopes 1.05, ECE 0.04 to 0.05 for the ensembles) but not the hard one (PSYKOSE $\rightarrow$ DEPRESJON: slopes 0.35 to 0.77, ECE 0.13 to 0.20), because the miscalibration there is a property of the population shift and cannot be estimated without data from the target population.

Table 9: E3: calibration alongside discrimination for every model in E1 (subject-wise arm) and E2, plus logistic recalibration of external predictions.

Exp.	Run	AUROC	Brier	Cal. slope	Cal. intercept	ECE
E1	depresjon.logreg.subject_wise	0.779	0.187	0.69	-0.51	0.102
E1	depresjon.rf.subject_wise	0.770	0.181	0.77	0.01	0.056
E1	depresjon.xgboost.subject_wise	0.764	0.192	0.53	0.27	0.105
E1	hyperaktiv.logreg.subject_wise	0.449	0.279	-0.28	-0.01	0.161
E1	hyperaktiv.rf.subject_wise	0.443	0.281	-0.45	0.00	0.137
E1	hyperaktiv.xgboost.subject_wise	0.442	0.313	-0.23	0.00	0.221
E1	psykose.logreg.subject_wise	0.907	0.112	0.95	-0.59	0.088
E1	psykose.rf.subject_wise	0.893	0.115	1.03	-0.12	0.052
E1	psykose.xgboost.subject_wise	0.903	0.103	0.78	0.14	0.052
E2	depresjon-to-hyperaktiv.logreg.external	0.442	0.350	-0.16	-0.42	0.291
E2	depresjon-to-hyperaktiv.rf.external	0.461	0.317	-0.15	0.33	0.214
E2	depresjon-to-hyperaktiv.xgboost.external	0.459	0.363	-0.09	0.47	0.292
E2	depresjon-to-psykose.logreg.external	0.779	0.193	0.50	0.06	0.079
E2	depresjon-to-psykose.rf.external	0.776	0.193	0.81	-0.02	0.044
E2	depresjon-to-psykose.xgboost.external	0.769	0.202	0.56	-0.01	0.085
E2	depresjon.logreg.internal	0.779	0.187	0.69	-0.51	0.102
E2	depresjon.rf.internal	0.770	0.181	0.77	0.01	0.056
E2	depresjon.xgboost.internal	0.764	0.192	0.53	0.27	0.105
E2	psykose-to-depresjon.logreg.external	0.689	0.262	0.27	1.20	0.202
E2	psykose-to-depresjon.rf.external	0.728	0.225	0.63	0.74	0.149
E2	psykose-to-depresjon.xgboost.external	0.671	0.284	0.31	1.43	0.229
E2	psykose.logreg.internal	0.885	0.137	0.77	-0.10	0.061
E2	psykose.rf.internal	0.854	0.151	0.82	-0.08	0.055
E2	psykose.xgboost.internal	0.877	0.148	0.62	-0.16	0.083
E2	depresjon-to-hyperaktiv.logreg.external+recal	0.442	0.311	-0.23	0.22	0.223
E2	depresjon-to-hyperaktiv.rf.external+recal	0.461	0.301	-0.20	0.37	0.176
E2	depresjon-to-hyperaktiv.xgboost.external+recal	0.459	0.311	-0.17	0.36	0.194
E2	depresjon-to-psykose.logreg.external+recal	0.779	0.211	0.72	0.56	0.139
E2	depresjon-to-psykose.rf.external+recal	0.776	0.193	1.05	0.11	0.041
E2	depresjon-to-psykose.xgboost.external+recal	0.769	0.196	1.05	0.11	0.052
E2	psykose-to-depresjon.logreg.external+recal	0.689	0.254	0.35	0.99	0.185
E2	psykose-to-depresjon.rf.external+recal	0.728	0.221	0.77	0.67	0.129
E2	psykose-to-depresjon.xgboost.external+recal	0.671	0.258	0.49	0.93	0.200

3.10 E4 and E5 results: sex strata; transdiagnostic and five-class tasks

Discrimination did not differ materially by sex on DEPRESJON (AUROC 0.76 to 0.78 in both strata) or on HYPERAKTIV (chance in both); on PSYKOSE the female stratum contains three cases and its apparently higher AUROC (0.97) rests on them, with intervals overlapping the male stratum (Table 10). Calibration slopes differed between strata more than discrimination did, which is the usual pattern when a model's probabilities are shifted for a subgroup that is under-represented in training. On OBF-Psychiatric, separating any psychiatric group from the healthy controls reached

Table 10: E4: sex-stratified performance of the subject-wise E1 models (window level, subject-level bootstrap intervals). Subgroups with fewer than six participants or a single class are omitted.

Cohort	Model	Sex	Participants (cases)	AUROC [95\% CI]	Brier	Cal. slope	ECE
depresjon	Logistic regression	female	30 (10)	0.777 [0.649, 0.872]	0.182	0.75	0.120
depresjon	Logistic regression	male	25 (13)	0.776 [0.562, 0.882]	0.192	0.62	0.091
depresjon	Random forest	female	30 (10)	0.763 [0.623, 0.870]	0.171	0.75	0.071
depresjon	Random forest	male	25 (13)	0.777 [0.626, 0.870]	0.193	0.80	0.070
depresjon	XGBoost	female	30 (10)	0.755 [0.620, 0.859]	0.183	0.51	0.109
depresjon	XGBoost	male	25 (13)	0.772 [0.619, 0.868]	0.203	0.55	0.120
hyperaktiv	Logistic regression	female	41 (21)	0.479 [0.370, 0.602]	0.275	-0.10	0.152
hyperaktiv	Logistic regression	male	44 (24)	0.424 [0.321, 0.543]	0.283	-0.49	0.168
hyperaktiv	Random forest	female	41 (21)	0.491 [0.371, 0.611]	0.269	-0.10	0.111
hyperaktiv	Random forest	male	44 (24)	0.398 [0.278, 0.516]	0.291	-0.80	0.160
hyperaktiv	XGBoost	female	41 (21)	0.498 [0.385, 0.605]	0.293	-0.02	0.211
hyperaktiv	XGBoost	male	44 (24)	0.388 [0.278, 0.505]	0.332	-0.44	0.230
psykose	Logistic regression	female	23 (3)	0.971 [0.915, 0.998]	0.096	1.58	0.177
psykose	Logistic regression	male	31 (19)	0.890 [0.771, 0.933]	0.126	0.83	0.081
psykose	Random forest	female	23 (3)	0.966 [0.891, 0.995]	0.075	1.66	0.147
psykose	Random forest	male	31 (19)	0.874 [0.765, 0.930]	0.149	0.90	0.123
psykose	XGBoost	female	23 (3)	0.969 [0.904, 0.997]	0.059	1.16	0.100
psykose	XGBoost	male	31 (19)	0.884 [0.774, 0.933]	0.140	0.71	0.121

Table 11: E5: OBF-Psychiatric (162 participants, single control group). Transdiagnostic binary arm (any psychiatric group vs healthy control; window-level AUROC with subject-level BCa intervals) and five-class arm (macro one-vs-rest AUROC), each under subject-wise and record-wise CV.

Arm	Model	Split	AUROC / macro-AUROC	Accuracy	Macro-F1	Cal. slope
Any psychiatric vs control	Majority class	subject-wise	0.490 [0.358, 0.614]	0.641	0.391	0.25
Any psychiatric vs control	Majority class	record-wise	0.500 [0.470, 0.529]	0.641	0.391	0.25
Any psychiatric vs control	Logistic regression	subject-wise	0.795 [0.745, 0.848]	0.741	0.725	0.78
Any psychiatric vs control	Logistic regression	record-wise	0.824 [0.777, 0.867]	0.756	0.742	0.93
Any psychiatric vs control	Random forest	subject-wise	0.781 [0.726, 0.837]	0.739	0.705	0.75
Any psychiatric vs control	Random forest	record-wise	0.849 [0.811, 0.887]	0.782	0.755	1.07
Any psychiatric vs control	XGBoost	subject-wise	0.776 [0.729, 0.834]	0.734	0.702	0.63
Any psychiatric vs control	XGBoost	record-wise	0.844 [0.810, 0.885]	0.781	0.755	0.90
Five diagnostic groups	Logistic regression	subject-wise	0.713 [0.677, 0.750]	0.431	0.390	–
Five diagnostic groups	Logistic regression	record-wise	0.770 [0.740, 0.799]	0.484	0.444	–
Five diagnostic groups	Random forest	subject-wise	0.695 [0.662, 0.735]	0.434	0.350	–
Five diagnostic groups	Random forest	record-wise	0.811 [0.786, 0.843]	0.558	0.482	–
Five diagnostic groups	XGBoost	subject-wise	0.702 [0.670, 0.743]	0.448	0.355	–
Five diagnostic groups	XGBoost	record-wise	0.808 [0.785, 0.842]	0.558	0.467	–

AUROC 0.776 to 0.795 subject-wise and 0.824 to 0.849 record-wise (participant level 0.856 to 0.867 versus 0.899 to 0.943); the five-class task gave a macro one-versus-rest AUROC of 0.695 to 0.713 (accuracy 0.43 to 0.45 against a 0.28 majority rate) subject-wise and 0.770 to 0.811 (0.48 to 0.56) record-wise (Table 11). Inflation was again largest for the ensembles (0.106 to 0.116 in macro-AUROC) and smallest for logistic regression (0.057). A transdiagnostic actigraphy signal exists, but it is moderate, and the five-way separation is weak once participants are held out.

3.11 E6 results: ablation

Honest performance was robust to the analytic choices a developer could plausibly vary (Table 12). On DEPRESJON, the full feature set gave AUROC 0.768 to 0.785 (logistic regression, XGBoost); dropping any single feature group changed it to 0.740 to 0.797; single groups alone ranged from 0.458 (day/night, logreg) to 0.802 (temporal/spectral, xgboost); SMOTE inside training folds gave 0.761 to 0.786 with ECE 0.097 to 0.109; leave-one-subject-out gave 0.745 to 0.773. On PSYKOSE, the full feature set gave AUROC 0.902 to 0.903 (logistic regression, XGBoost); dropping any single feature group changed it to 0.845 to 0.901; single groups alone ranged from 0.703 (day/night, logreg) to 0.875 (distributional, logreg); SMOTE inside training folds gave 0.904 to 0.906 with ECE 0.046 to 0.082; leave-one-subject-out gave 0.899 to 0.906. On HYPERAKTIV, the full feature set gave AUROC 0.452 to 0.466 (logistic regression, XGBoost); dropping any single feature group changed it to 0.418 to 0.493; single groups alone ranged from 0.459 (circadian, xgboost) to 0.519 (temporal/spectral, logreg); SMOTE inside training folds gave 0.454 to 0.466 with ECE 0.149 to 0.208; leave-one-subject-out gave 0.424 to 0.472. No variant on HYPERAKTIV rose above chance under subject-wise evaluation, and no variant on DEPRESJON or PSYKOSE approached the record-wise estimates of E1: the gap between published and honest numbers is a property of the splitting design, not of feature engineering, resampling or the cross-validation scheme.

3.12 Published claims against the measured bands

Table 13 places the Part I claims beside the Part II bands. Every published accuracy above the subject-wise band on DEPRESJON (0.89, 0.96, 0.85) lies inside or above the record-wise band, and every such study used, or did not state, a day-level unit; the two PSYKOSE claims of 0.92 and 0.94 sit at the top of the record-wise band. The honest participant-level AUROC of 0.87 to 0.90 for depression agrees with the dataset authors' leave-one-patient-out baseline [1] and is what a new participant should expect; the leaky 0.97 to 0.99 is what the literature reports. For ADHD versus clinical controls the honest band is chance and no published claim above it survives subject-wise evaluation. The comparison is not a criticism of any individual study: the numbers reproduce. It is a statement about what they measure.

4 Discussion

4.1 Principal findings

Each practice the appraisal flags changed conclusions on real data, always towards optimism. Record-wise splitting added 0.02 to 0.17 AUROC and turned a null result into an apparently useful ADHD classifier. Skipping external validation hid a loss of up to 0.21 AUROC and a collapse of calibration in one transfer direction while the other direction happened to hold, which is why transfer must be tested rather than assumed. Calibration, unreported in the entire literature on these cohorts, was poor under honest validation and was not repairable from the training data alone. The honest results were robust to feature subsets, resampling inside folds and the cross-validation scheme, so the gap between published and honest numbers is a property of the splitting design and not of feature engineering.

Table 12: E6: ablation under subject-wise CV. Feature-group ablations (drop-one and only-one), SMOTE applied inside training folds, and leave-one-subject-out (LOSO) in place of five-fold CV. Window-level estimates with subject-level BCa intervals.

Cohort	Model	Variant	k	AUROC [95% CI]	Subject AUROC	Brier	Cal. slope	ECE
depresjon	Logistic regression	all_features	28	0.785 [0.694, 0.853]	0.902	0.187	0.71	0.103
depresjon	Logistic regression	drop_distributional	16	0.758 [0.649, 0.837]	0.890	0.198	0.69	0.119
depresjon	Logistic regression	only_distributional	12	0.755 [0.652, 0.836]	0.838	0.200	0.79	0.115
depresjon	Logistic regression	drop_day_night	24	0.781 [0.687, 0.850]	0.893	0.188	0.74	0.108
depresjon	Logistic regression	only_day_night	4	0.458 [0.358, 0.541]	0.577	0.257	-0.07	0.149
depresjon	Logistic regression	drop_circadian	22	0.797 [0.708, 0.863]	0.906	0.181	0.78	0.103
depresjon	Logistic regression	only_circadian	6	0.640 [0.533, 0.722]	0.783	0.239	0.59	0.152
depresjon	Logistic regression	drop_temporal_spectral	22	0.771 [0.678, 0.840]	0.890	0.193	0.73	0.102
depresjon	Logistic regression	only_temporal_spectral	6	0.740 [0.551, 0.829]	0.874	0.211	0.85	0.178
depresjon	Logistic regression	all_features+smote_in_fold	28	0.786 [0.697, 0.852]	0.899	0.186	0.68	0.097
depresjon	Logistic regression	all_features+loso	28	0.773 [0.660, 0.849]	0.890	0.191	0.63	0.111
depresjon	XGBoost	all_features	28	0.768 [0.685, 0.837]	0.871	0.192	0.53	0.101
depresjon	XGBoost	drop_distributional	16	0.788 [0.711, 0.852]	0.905	0.181	0.59	0.093
depresjon	XGBoost	only_distributional	12	0.739 [0.640, 0.812]	0.844	0.196	0.54	0.090
depresjon	XGBoost	drop_day_night	24	0.769 [0.685, 0.836]	0.878	0.190	0.53	0.093
depresjon	XGBoost	only_day_night	4	0.769 [0.682, 0.841]	0.875	0.185	0.63	0.062
depresjon	XGBoost	drop_circadian	22	0.787 [0.707, 0.851]	0.912	0.181	0.58	0.090
depresjon	XGBoost	only_circadian	6	0.711 [0.606, 0.796]	0.817	0.203	0.52	0.090
depresjon	XGBoost	drop_temporal_spectral	22	0.740 [0.642, 0.817]	0.825	0.204	0.49	0.117
depresjon	XGBoost	only_temporal_spectral	6	0.802 [0.737, 0.859]	0.947	0.168	0.68	0.053
depresjon	XGBoost	all_features+smote_in_fold	28	0.761 [0.676, 0.835]	0.879	0.196	0.51	0.109
depresjon	XGBoost	all_features+loso	28	0.745 [0.649, 0.820]	0.853	0.205	0.45	0.124
psykose	Logistic regression	all_features	28	0.903 [0.807, 0.937]	0.973	0.115	0.92	0.090
psykose	Logistic regression	drop_distributional	16	0.845 [0.754, 0.892]	0.952	0.159	0.87	0.119
psykose	Logistic regression	only_distributional	12	0.875 [0.785, 0.921]	0.945	0.136	0.95	0.099
psykose	Logistic regression	drop_day_night	24	0.890 [0.825, 0.926]	0.979	0.125	0.94	0.083
psykose	Logistic regression	only_day_night	4	0.703 [0.575, 0.791]	0.852	0.223	0.74	0.195
psykose	Logistic regression	drop_circadian	22	0.890 [0.810, 0.932]	0.967	0.125	0.92	0.092
psykose	Logistic regression	only_circadian	6	0.784 [0.714, 0.839]	0.926	0.192	0.85	0.119
psykose	Logistic regression	drop_temporal_spectral	22	0.900 [0.810, 0.936]	0.969	0.117	0.95	0.095
psykose	Logistic regression	only_temporal_spectral	6	0.733 [0.665, 0.790]	0.862	0.212	0.94	0.146
psykose	Logistic regression	all_features+smote_in_fold	28	0.906 [0.810, 0.939]	0.973	0.113	0.86	0.082
psykose	Logistic regression	all_features+loso	28	0.906 [0.811, 0.939]	0.974	0.114	0.92	0.092
psykose	XGBoost	all_features	28	0.902 [0.806, 0.939]	0.969	0.104	0.77	0.045
psykose	XGBoost	drop_distributional	16	0.901 [0.813, 0.937]	0.969	0.103	0.79	0.064
psykose	XGBoost	only_distributional	12	0.874 [0.768, 0.920]	0.950	0.124	0.74	0.054
psykose	XGBoost	drop_day_night	24	0.887 [0.793, 0.926]	0.956	0.114	0.78	0.061
psykose	XGBoost	only_day_night	4	0.854 [0.739, 0.903]	0.956	0.134	0.73	0.059
psykose	XGBoost	drop_circadian	22	0.897 [0.796, 0.937]	0.967	0.105	0.76	0.052
psykose	XGBoost	only_circadian	6	0.843 [0.763, 0.889]	0.935	0.143	0.77	0.060
psykose	XGBoost	drop_temporal_spectral	22	0.900 [0.807, 0.935]	0.963	0.108	0.77	0.045
psykose	XGBoost	only_temporal_spectral	6	0.854 [0.781, 0.905]	0.969	0.135	0.75	0.046
psykose	XGBoost	all_features+smote_in_fold	28	0.904 [0.811, 0.940]	0.972	0.106	0.76	0.046
psykose	XGBoost	all_features+loso	28	0.899 [0.810, 0.936]	0.970	0.109	0.73	0.051
hyperaktiv	Logistic regression	all_features	28	0.466 [0.383, 0.554]	0.456	0.275	-0.17	0.157
hyperaktiv	Logistic regression	drop_distributional	16	0.472 [0.401, 0.557]	0.459	0.267	-0.15	0.130
hyperaktiv	Logistic regression	only_distributional	12	0.461 [0.364, 0.544]	0.426	0.264	-0.37	0.082
hyperaktiv	Logistic regression	drop_day_night	24	0.476 [0.394, 0.556]	0.455	0.270	-0.08	0.133
hyperaktiv	Logistic regression	only_day_night	4	0.493 [0.376, 0.594]	0.471	0.258	-0.40	0.047
hyperaktiv	Logistic regression	drop_circadian	22	0.418 [0.323, 0.510]	0.391	0.277	-0.52	0.133
hyperaktiv	Logistic regression	only_circadian	6	0.486 [0.398, 0.558]	0.495	0.259	-0.30	0.051
hyperaktiv	Logistic regression	drop_temporal_spectral	22	0.493 [0.406, 0.577]	0.490	0.267	-0.07	0.130
hyperaktiv	Logistic regression	only_temporal_spectral	6	0.519 [0.418, 0.612]	0.505	0.255	0.05	0.028
hyperaktiv	Logistic regression	all_features+smote_in_fold	28	0.466 [0.383, 0.555]	0.453	0.275	-0.17	0.149
hyperaktiv	Logistic regression	all_features+loso	28	0.472 [0.388, 0.553]	0.457	0.275	-0.16	0.146
hyperaktiv	XGBoost	all_features	28	0.452 [0.378, 0.548]	0.444	0.311	-0.18	0.207
hyperaktiv	XGBoost	drop_distributional	16	0.453 [0.383, 0.535]	0.451	0.307	-0.17	0.207
hyperaktiv	XGBoost	only_distributional	12	0.517 [0.442, 0.594]	0.536	0.287	0.03	0.155
hyperaktiv	XGBoost	drop_day_night	24	0.467 [0.398, 0.562]	0.477	0.304	-0.11	0.198
hyperaktiv	XGBoost	only_day_night	4	0.493 [0.424, 0.573]	0.504	0.288	-0.07	0.154
hyperaktiv	XGBoost	drop_circadian	22	0.454 [0.373, 0.540]	0.445	0.306	-0.16	0.210
hyperaktiv	XGBoost	only_circadian	6	0.459 [0.398, 0.524]	0.461	0.295	-0.20	0.171
hyperaktiv	XGBoost	drop_temporal_spectral	22	0.471 [0.401, 0.565]	0.467	0.304	-0.14	0.195
hyperaktiv	XGBoost	only_temporal_spectral	6	0.470 [0.407, 0.538]	0.460	0.292	-0.09	0.174
hyperaktiv	XGBoost	all_features+smote_in_fold	28	0.454 [0.379, 0.543]	0.453	0.310	-0.18	0.208
hyperaktiv	XGBoost	all_features+loso	28	0.424 [0.344, 0.503]	0.399	0.334	-0.25	0.275

Table 13: Published best accuracies on each cohort (Part I seed list; Table 1) against the accuracy and AUROC bands measured in Part II under subject-wise and record-wise five-fold cross-validation (window level unless stated; seed-averaged estimates, range over logistic regression, random forest and XGBoost).

Cohort	Published best accuracies (ID value)	Part I accuracy, subject-wise (range over models)	Part II accuracy, record-wise	Part I AUROC, subject-wise	Part II AUROC, record-wise	Part II subject-level AUROC, subject-wise
DEPRESJON	S02 0.89, S07 0.77, S09 0.78, S10 0.96, S16 0.94, S19 0.85	0.72–0.72	0.77–0.81	0.76–0.78	0.84–0.88	0.87–0.90
PSYKOSE	S15 0.92, S16 0.94	0.86–0.87	0.87–0.89	0.89–0.91	0.93–0.95	0.96–0.97
HYPERAKTIV	–	0.44–0.46	0.53–0.57	0.44–0.45	0.58–0.61	0.42–0.44

4.2 Mechanism

Why does record-wise splitting inflate so much here? Each participant contributes 6 to 16 days, and a day’s activity profile is strongly participant-specific: habitual wake time, weekday routine, overall activity level and rest fragmentation vary far more between people than a disorder shifts them within a person. A flexible model trained on some of a participant’s days and tested on others can match the test day to its owner and read off the owner’s label. The evidence is direct: inflation grew with model flexibility, it was largest where the disorder signal was weakest (HYPERAKTIV), and the convolutional network, which can encode the full daily profile, showed the largest participant-level jump. The same mechanism explains why record-wise models look well calibrated: their confidence is warranted for the participants they memorised. None of this is specific to actigraphy; it applies to any repeated-measures data, and the PROBAST+AI signalling question on data leakage exists for this reason.

4.3 Why the combination matters

An appraisal alone counts practices; a re-analysis alone is one more model on familiar data. Together, on the same cohorts, they show that the practices the appraisal flags are the practices that produce the reported numbers. This is the strongest form of evidence a methodological critique can offer: not that a practice *could* inflate results, but that it *did*, by how much, and for which claims. It also gives the field something it lacks: a reference band. Future work on these cohorts can be read against Tables 3, 8 and 13, and a claim above the honest band invites one question, namely which splitting unit produced it.

4.4 Clinical implications

For a screening application the relevant quantities are participant-level discrimination and calibration on people the model has never seen, in the population where it will be used. On these data, those quantities are an AUROC of about 0.87 to 0.90 for depression and 0.96 to 0.97 for schizophrenia against healthy controls, no discrimination for ADHD against other psychiatric outpatients, and probabilities that need recalibration in the target population before they can inform a decision. The distinction between healthy and clinical controls matters clinically: a model that separates depressed patients from healthy volunteers has not been shown to separate them from the anxious, sleep-disordered or otherwise unwell people who make up a referral population, and HYPERAKTIV’s clinical controls show how quickly the actigraphy signal disappears when the comparison becomes realistic.

4.5 Limitations

One device family and one country; small participant numbers with wide participant-level intervals; diagnostic-group labels rather than symptom trajectories; one window length; models fixed rather than tuned (tuning inside subject-wise folds would raise honest performance somewhat without changing the paired inflation, which is the estimand); no demographic predictors; in E2, a control group split rather than an independent one, because no independent healthy-control actigraphy with the same device is public; the PSYKOSE sex analysis is underpowered; and Part I is a closed-scope appraisal identified by citation tracing and dataset-name searches rather than exhaustive licensed-database searches, was not prospectively registered, and rests on abstracts and metadata for the studies marked in Table 1, so the unit of splitting may be understated for some of them (an unstated unit was never counted as day-level). We did not attempt to reproduce individual published pipelines; the bands are for fixed, conventional models, and a specific published architecture could sit above them for reasons other than leakage, which its authors can now demonstrate by reporting subject-wise results.

4.6 Recommendations and conclusion

For any repeated-measures mental-health dataset: state the unit of splitting; report participant-level as well as window-level sample sizes and metrics; report calibration slope and intercept with a reliability curve; check for shared participants across combined datasets; release code with run manifests. Journals could require each in one sentence. This repository provides a reference implementation and honest baselines for the three cohorts; claims that exceed them should be accompanied by the unit of splitting that produced them.

Declarations

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Supplementary material

A single supplementary document ([supplementary.pdf](#)) accompanies this article: Table S1, TRI-POD+AI checklist; Table S2, feature definitions; Table S3, identification sources for Part I; Table S4, study sources and extraction basis; Table S5, run-manifest fields and the per-window prediction files; Figures S1 to S3, ROC and reliability curves for every cohort and cohort pair.

References

- [1] Garcia-Ceja E, Riegler M, Jakobsen P, et al. Depresjon: a motor activity database of depression episodes in unipolar and bipolar patients. ACM MMSys. 2018. doi:10.1145/3204949.3208125
- [2] Jakobsen P, Garcia-Ceja E, Stabell LA, et al. PSYKOSE: a motor activity database of patients with schizophrenia. IEEE CBMS. 2020. doi:10.1109/CBMS49503.2020.00064
- [3] Hicks SA, Stautland A, Fasmer OB, et al. HYPERAKTIV: an activity dataset from patients with attention-deficit/hyperactivity disorder (ADHD). ACM MMSys. 2021. doi:10.1145/3458305.3478454

- [4] Garcia-Ceja E, et al. OBF-Psychiatric, a motor activity dataset of patients diagnosed with major depression, schizophrenia, and ADHD. *Sci Data*. 2025;12. doi:10.1038/s41597-025-04384-3
- [5] Explainable AI for depression detection and severity classification from activity data. *JMIR Ment Health*. 2025;12:e72038.
- [6] Transfer learning for real-time deployment of a screening tool for depression detection using actigraphy. *arXiv:2303.07847*. 2023.
- [7] Zanella-Calzada LA, et al. Feature extraction in motor activity signal: towards a depression episodes detection in unipolar and bipolar patients. *Diagnostics*. 2019;9:8.
- [8] Objective diagnosis of psychotic disorders using multi-branch deep learning on time-series motor activity signals. *Soft Comput*. 2024. doi:10.1007/s00500-024-09832-7
- [9] Unveiling psychotic disorder patterns: a deep learning model analysing motor activity time-series data with explainable AI. *Biomed Signal Process Control*. 2024.
- [10] Saeb S, Lonini L, Jayaraman A, Mohr DC, Kording KP. The need to approximate the use-case in clinical machine learning. *GigaScience*. 2017;6:gix019.
- [11] Karbalaie A, Abtahi F, Häger C. Participant-aware model validation for repeated-measures data: comparative cross-validation study. *JMIR AI*. 2026;5:e87728.
- [12] Van Calster B, McLernon DJ, van Smeden M, et al. Calibration: the Achilles heel of predictive analytics. *BMC Med*. 2019;17:230.
- [13] Moons KGM, Damen JAA, Nair T, et al. PROBAST+AI: an updated quality, risk of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods. *BMJ*. 2025;388:e082505.
- [14] Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*. 2024;385:e078378.
- [15] Meehan AJ, Lewis SJ, Fazel S, et al. Clinical prediction models in psychiatry: a systematic review of two decades of progress and challenges. *Mol Psychiatry*. 2022;27:2700–2708.